# Notes on Algerbra

Shorya Prakash

October 17, 2026

## Preface

These notes are motivated by the Algebra course taken by the author at the Institute of Mathematical Sciences (IMSc), Chennai, taught by Prof. Sankaran Vishwanath.

The purpose of these notes is to organise and develop the material covered in the course in a form suitable for future reference and study. These notes are not suitable for people studying group theory or rings and modules for the first time, since many arguments might be more hand-wavy for a first-time reader. However, the author believes these arguments contain enough information that people can complete them when required.

The preparation of this document involved substantial use of artificial intelligence tools for writing and exposition. However, the author has reviewed and verified everything to the best of their ability. Despite this effort, errors or inaccuracies may remain, and the notes should not be regarded as error-free.

# Table of Contents

Monoids ..... *To be added*

## 0.1 Monoids

**Definition 0.1** (Monoid)**.** A **monoid** is a pair $(M, \cdot)$ consisting of a set $M$ and a binary operation $\cdot : M \times M \to M$ satisfying the following two axioms:

1. **Associativity:** For all $a, b, c \in M$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
2. **Identity Element:** There exists an element $1_M \in M$ such that for all $m \in M$, $1_M \cdot m = m \cdot 1_M = m$.

## 0.2 Examples of Monoids

(a) The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ forms a monoid under multiplication, where the identity element is 1.

(b) The set $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ forms a monoid under addition, where the identity element is 0.

(c) Any group $(G, *)$ is naturally a monoid, as it satisfies associativity and possesses an identity element.

(d) If $R$ is a ring with identity, then $R$ forms a monoid under its multiplication operation. Specifically, the set of $n \times n$ matrices over a ring with identity, denoted $M_n(R)$, forms a monoid under matrix multiplication, where the identity element is the identity matrix $I_n$.

(e) The set of functions from X to X forms a monoid under composition.

**Definition 0.2** (Monoid Homomorphism )**.** Let $M$ and $N$ be monoids. A map $\varphi : M \to N$ is a **monoid homomorphism** if:

1. $\varphi(m_1 m_2) = \varphi(m_1)\varphi(m_2)$ for all $m_1, m_2 \in M$,
2. $\varphi(1_M) = 1_N$.

If a monoid homomorphism $\varphi$ is bijective, it is called a **monoid isomorphism**. Two monoids $M$ and $N$ are said to be **isomorphic** (denoted $M \cong N$) if there exists a monoid isomorphism between them.

**Exercise 0.1.** Let $\mathcal{M}$ be the class of all monoids. Define a relation $\sim$ on $\mathcal{M}$ such that $M \sim N$ if and only if there exists a monoid isomorphism from $M$ to $N$.

(a) Explain why $\mathcal{M}$ is a class and clearly not a set (such a class is called a **proper class**).
*Hint: Russell's Paradox*

(b) Prove that $\sim$ is an equivalence relation on $\mathcal{M}$.

**Exercise 0.2.** Prove that the monoids $(\mathbb{N}, \cdot)$ and $(\mathbb{N}_0, +)$ given in Example 1.1(a) and 1.1(b) are not isomorphic.

**Exercise 0.3.** Prove that the product of $n$ monoid elements $a_1, a_2, \dots, a_n$ taken in that exact order is independent of how the brackets are placed.
*Hint: Use strong induction on $n$.*

## 0.3 The Free Monoid

**Definition 0.3** (Free Monoid)**.** Let $X$ be an arbitrary set (often called an *alphabet*). The **free monoid** on $X$, denoted by $X^*$, is the set of all finite words (strings) formed by elements of $X$:

$$X^* = \{x_1 x_2 \cdots x_n \mid n \geq 0,\, x_i \in X\}.$$

When $n = 0$, the word is the empty word, denoted by 1 (or $e$).
Equivalently, we can construct $X^*$ in terms of Cartesian products. Let $X_0 = \{1\}$, where 1 (or $e$) denotes the empty word of length 0. For each $n \geq 1$, let $X_n$ denote the $n$-fold Cartesian product:

$$X_n = \underbrace{X \times X \times \cdots \times X}_{n \text{ times}} = \{(x_1, x_2, \ldots, x_n) \mid x_i \in X\}.$$

Identifying the ordered $n$-tuple $(x_1, x_2, \ldots, x_n) \in X_n$ with the concatenated word $x_1 x_2 \cdots x_n$, the free monoid is the disjoint union (where is such disjoint union defined):

$$X^* = \bigcup_{n=0}^{\infty} X_n.$$

The binary operation on $X^*$ is string concatenation:

$$(x_1 \cdots x_n) \cdot (y_1 \cdots y_m) = x_1 \cdots x_n y_1 \cdots y_m,$$

which is associative, with the empty word 1 serving as the two-sided identity.

**Example 0.1** (Free Monoid on One Generator)**.** Let $X = \{x\}$ be a singleton set. Every word in $X^*$ consists solely of repeated concatenations of the single letter $x$:

$$X^* = \{1, x, x^2, x^3, \ldots\} = \{x^n \mid n \geq 0\},$$

where $x^0 = 1$ denotes the empty word. The concatenation operation obeys:

$$x^m \cdot x^n = x^{m+n} \quad \text{for all } m, n \geq 0.$$

In particular, $\{x\}^*$ is commutative. The map

$$\psi : \{x\}^* \longrightarrow (\mathbb{N}_0, +), \quad \psi(x^n) = n$$

is a bijection satisfying $\psi(x^m \cdot x^n) = m + n = \psi(x^m) + \psi(x^n)$ and $\psi(1) = 0$. Hence, the free monoid on one generator is isomorphic to the additive monoid of non-negative integers:

$$\{x\}^* \cong (\mathbb{N}_0, +).$$

**Example 0.2** (Free Monoid on Two Generators)**.** Let $X = \{a, b\}$. The elements of $X^*$ are all finite strings formed from the two-letter alphabet $\{a, b\}$. Grouped systematically by word length $n$:

- **Length 0:** 1 (the empty word, $2^0 = 1$ element)
- **Length 1:** $a, b$ ($2^1 = 2$ elements)

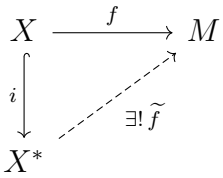
- **Length 2:** $a^2, ab, ba, b^2$ ($2^2 = 4$ elements)
- **Length 3:** $a^3, a^2b, aba, ab^2, ba^2, bab, b^2a, b^3$ ($2^3 = 8$ elements)

In general, there are $2^n$ distinct words of length $n$, growing exponentially.

**Theorem 0.1** (Universal Property of Free Monoids)**.** *Let $X$ be an arbitrary set, and let $i : X \hookrightarrow X^*$ be the canonical inclusion map sending $x \mapsto x$ (viewed as a word of length 1).*

*For every monoid $(M, \cdot)$ and any set map $f : X \to M$, there exists a unique monoid homomorphism $\widetilde{f} : X^* \to M$ such that $\widetilde{f} \circ i = f$.*

*That is, the following diagram commutes:*




*Proof.* **Existence:** Let $w \in X^*$. If $w = 1$ (the empty word), we define $\widetilde{f}(1) = 1_M$. If $w$ is a non-empty word, it can be expressed uniquely as a finite sequence of letters:

$$w = x_1 x_2 \cdots x_k, \quad \text{where } k \geq 1 \text{ and } x_j \in X.$$

We define:

$$\widetilde{f}(w) = f(x_1) \cdot f(x_2) \cdots f(x_k) \in M.$$

To verify that $\widetilde{f}$ is a monoid homomorphism:

(i) *Operation preservation:* Let $u = x_1 \cdots x_k$ and $v = y_1 \cdots y_m$ be two non-empty words in $X^*$. Then $u \cdot v = x_1 \cdots x_k y_1 \cdots y_m$, and by associativity in $M$:

$$\widetilde{f}(u \cdot v) = f(x_1) \cdots f(x_k) f(y_1) \cdots f(y_m) = \widetilde{f}(u) \cdot \widetilde{f}(v).$$

If either $u = 1$ or $v = 1$, this identity holds trivially.

(ii) *Identity preservation:* By construction, $\widetilde{f}(1) = 1_M$.

Furthermore, for any letter $x \in X$, $i(x) = x$ is a word of length 1, so:

$$(\widetilde{f} \circ i)(x) = \widetilde{f}(x) = f(x).$$

Thus $\widetilde{f} \circ i = f$.

**Uniqueness:** Suppose $g : X^* \to M$ is another monoid homomorphism satisfying $g \circ i = f$. Because $g$ is a homomorphism, $g(1) = 1_M = \widetilde{f}(1)$. For any word $w = x_1 x_2 \cdots x_k \in X^*$:

$$g(w) = g(x_1 x_2 \cdots x_k) = g(x_1) g(x_2) \cdots g(x_k).$$

Since $g \circ i = f$, we have $g(x_j) = g(i(x_j)) = f(x_j)$ for all $1 \leq j \leq k$. Hence:

$$g(w) = f(x_1) f(x_2) \cdots f(x_k) = \widetilde{f}(w).$$

Therefore, $g = \widetilde{f}$, establishing uniqueness. □

**Proposition 0.1** (Properties of the Free Monoid)**.** *Let $X$ be an arbitrary set and $X^*$ the free monoid on $X$.*

(a) ***Cancellation Laws:*** *The free monoid $X^*$ satisfies both left and right cancellation: for all words $u, v, w \in X^*$,*

$$wu = wv \implies u = v \quad \textit{and} \quad uw = vw \implies u = v.$$

(b) ***Trivial Units:*** *The group (why?) of units (invertible elements) of $X^*$ is trivial:*

$$(X^*)^\times = \{e\}.$$

*Infact, if $u, v \in X^*$ satisfy $u \cdot v = e$, then $u = v = e$.*

(c) ***Characterization of Commuting Words:*** *Two words $u, v \in X^*$ commute ($uv = vu$) if and only if they are integer powers of a common word:*

$$uv = vu \iff \exists\, w \in X^* \textit{ and } k, m \geq 0 \textit{ such that } u = w^k \textit{ and } v = w^m.$$

*Consequently, $X^*$ is commutative if and only if $|X| \leq 1$. If $|X| \geq 2$, choosing distinct letters $a \neq b \in X$ yields $ab \neq ba$, so $X^*$ is non-commutative*

*Proof.* Let $\ell : X \to (\mathbb{N}_0, +)$ denote the word length function defined by $\ell(x) = 1$ for $x \in X$. This induces a monoid homomorphism (abuse of notation) $\ell : X^* \to (\mathbb{N}_0, +)$ satisfying $\ell(u \cdot v) = \ell(u) + \ell(v)$ and $\ell(e) = 0$. Verify $\ell(x) = 0 \iff x = e$

(a) *Cancellation:*

(b) *Units:* If $u \cdot v = e$, then $\ell(u \cdot v) = \ell(e) = 0$, so $\ell(u) + \ell(v) = 0$. This forces $\ell(u) = \ell(v) = 0$, which implies $u = v = e$.

(c) *Commutativity:*

□

**Example 0.3** (Failure of Free Monoid Properties in General Monoids)**.** The properties in Proposition 1.1 highlight the rigid nature of free monoids. General monoids can violate any three conditions (a), (b), and (c).

Consider the integers under multiplication, $(\mathbb{Z}, \cdot)$, with identity element 1:

(a) **Failure of Cancellation:** In $(\mathbb{Z}, \cdot)$, both left and right cancellation fail due to the zero element:

$$0 \cdot 3 = 0 \cdot 5 = 0, \quad \text{yet } 3 \neq 5.$$

Thus, $wu = wv$ does not imply $u = v$.

(b) **Failure of Trivial Units:** The group of invertible elements (units) in $(\mathbb{Z}, \cdot)$ is:

$$(\mathbb{Z}, \cdot)^\times = \{1, -1\}.$$

(c) **Failure of the Common-Power Commutativity Characterization:** The monoid $(\mathbb{Z}, \cdot)$ is commutative, so every pair of elements commutes:

$$3 \cdot 5 = 5 \cdot 3 = 15.$$

However, 3 and 5 are distinct prime numbers; there is no common base integer $w \in \mathbb{Z}$ and powers $k, m \geq 0$ such that $3 = w^k$ and $5 = w^m$.

**Exercise 0.4.** Find a monoid that is not a free monoid but satisfies properties (a), (b) and (c) of proposition 0.1. Find particularly a non-abelian monoid.

**Exercise 0.5** (Uniqueness of Free Monoids via the Universal Property)**.** Suppose $M$ is a monoid containing $X$ as a subset such that the triple $(M, X, i)$ also satisfies the universal property of free monoids, where $i : X \hookrightarrow M$ is the inclusion map. That is, for every monoid $N$ and every set map $f : X \to N$, there exists a unique monoid homomorphism $\widetilde{f} : M \to N$ such that $\widetilde{f} \circ i = f$.

Prove that $M$ is isomorphic to $X^*$ ($M \cong X^*$), and that this isomorphism preserves the generating set $X$ pointwise.

## 0.4 Congruences and Quotient Monoids

**Definition 0.4** (Multiplicative Equivalence Relation on Monoid)**.** Let $(M, \cdot)$ be a monoid. An equivalence relation $\sim$ on $M$ is called a **multiplicative equivalence relation** (or a **monoid congruence**) if for all $w_1, w_2, w_1', w_2' \in M$:

$$w_1 \sim w_2 \quad \text{and} \quad w_1' \sim w_2' \implies w_1 w_1' \sim w_2 w_2'.$$

(Verify!) Equivalently, $\sim$ is a congruence if and only if it is compatible with multiplication on both sides:

$$a \sim b \implies ca \sim cb \quad \text{and} \quad ac \sim bc \quad \text{for all } a, b, c \in M.$$

**Theorem 0.2** (Quotient Monoid)**.** *Let $(M, \cdot)$ be a monoid and let $\sim$ be an equivalence relation on $M$. Let $M/\sim = \{[x] \mid x \in M\}$ denote the set of equivalence classes under $\sim$.*

*Define the binary operation on $M/\sim$ by:*

$$[x] \cdot [y] := [xy] \quad \textit{for all } x, y \in M.$$

*Then:*

*(a) The operation $[x] \cdot [y] = [xy]$ is well-defined if and only if $\sim$ is a congruence on $M$.*

*(b) When $\sim$ is a congruence, $(M/\sim, \cdot)$ forms a monoid with identity element $[1_M]$, called the **quotient monoid**.*

*Proof.* **Part (a): Necessity and Sufficiency of Well-Definedness.**

- ($\implies$ *Necessity*): Suppose the operation $[x] \cdot [y] = [xy]$ is well-defined on $M/\sim$. Let $w_1, w_2, w_1', w_2' \in M$ such that $w_1 \sim w_2$ and $w_1' \sim w_2'$.

  By the definition of equivalence classes, $[w_1] = [w_2]$ and $[w_1'] = [w_2']$. Since the operation is well-defined, the product of classes is independent of the chosen representatives:

  $$[w_1 w_1'] = [w_1] \cdot [w_1'] = [w_2] \cdot [w_2'] = [w_2 w_2'].$$

  Thus $[w_1 w_1'] = [w_2 w_2']$, which means $w_1 w_1' \sim w_2 w_2'$. Hence, $\sim$ must be a congruence.

- ($\impliedby$ *Sufficiency*): Suppose $\sim$ is a congruence on $M$. To show that the operation is well-defined, let $x, x', y, y' \in M$ such that $[x] = [x']$ and $[y] = [y']$. Then $x \sim x'$ and $y \sim y'$. By the definition of a congruence:

  $$xy \sim x'y',$$

  which implies $[xy] = [x'y']$. Thus the product $[x] \cdot [y] = [xy]$ does not depend on the choice of representatives $x \in [x]$ and $y \in [y]$, so the operation is well-defined.

**Part (b): Verification of Monoid Axioms.** Assuming $\sim$ is a congruence:

(i) **Associativity:** For all $[x], [y], [z] \in M/\sim$, using the associativity of multiplication in $M$:

$$([x] \cdot [y]) \cdot [z] = [xy] \cdot [z] = [(xy)z] = [x(yz)] = [x] \cdot [yz] = [x] \cdot ([y] \cdot [z]).$$

(ii) **Identity Element:** The class $[1_M]$ of the identity element $1_M \in M$ serves as the two-sided identity: for all $[x] \in M/\sim$,

$$[1_M] \cdot [x] = [1_M x] = [x] \quad \text{and} \quad [x] \cdot [1_M] = [x 1_M] = [x].$$

Hence, $(M/\sim, \cdot)$ satisfies both monoid axioms and is a monoid. □

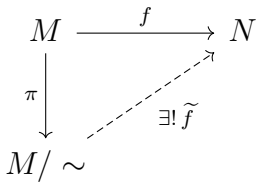
**Theorem 0.3** (Universal Property of the Quotient Monoid)**.** *Let $M$ be a monoid, $\sim$ a congruence on $M$, and $\pi : M \to M/\sim$ the canonical projection map defined by $\pi(x) = [x]$.*

*The map $\pi$ is a surjective monoid homomorphism. Furthermore, for every monoid $N$ and every monoid homomorphism $f : M \to N$ satisfying:*

$$x \sim y \implies f(x) = f(y) \quad \textit{for all } x, y \in M,$$

*there exists a* ***unique*** *monoid homomorphism $\widetilde{f} : M/\sim \to N$ such that $\widetilde{f} \circ \pi = f$.*

*That is, the following diagram commutes:*



*Proof.* First, $\pi$ is a monoid homomorphism because:

$$\pi(xy) = [xy] = [x] \cdot [y] = \pi(x) \cdot \pi(y), \quad \text{and} \quad \pi(1_M) = [1_M] = 1_{M/\sim}.$$

Surjectivity is immediate since every class $[x] \in M/\sim$ is the image $\pi(x)$.

**Existence:** For any class $[x] \in M/\sim$, define:

$$\widetilde{f}([x]) := f(x).$$

- *Well-definedness:* If $[x] = [y]$, then $x \sim y$. By the given condition on $f$, this implies $f(x) = f(y)$. Hence $\widetilde{f}$ is well-defined.
- *Homomorphism properties:* Because $f$ is a homomorphism:

  $$\widetilde{f}([x] \cdot [y]) = \widetilde{f}([xy]) = f(xy) = f(x)f(y) = \widetilde{f}([x]) \cdot \widetilde{f}([y]),$$

  and

  $$\widetilde{f}(1_{M/\sim}) = \widetilde{f}([1_M]) = f(1_M) = 1_N.$$

  Thus $\widetilde{f}$ is a monoid homomorphism.
- *Commutativity of diagram:* For all $x \in M$, $(\widetilde{f} \circ \pi)(x) = \widetilde{f}(\pi(x)) = \widetilde{f}([x]) = f(x)$, so $\widetilde{f} \circ \pi = f$.

**Uniqueness:** Suppose $g : M/\sim \to N$ is another monoid homomorphism satisfying $g \circ \pi = f$. For any class $[x] \in M/\sim$:

$$g([x]) = g(\pi(x)) = (g \circ \pi)(x) = f(x) = \widetilde{f}([x]).$$

Hence $g = \widetilde{f}$. □

**Exercise 0.6** (Congruences on Groups)**.** Let $(G, \cdot)$ be a group, and let $\sim$ be an equivalence relation on $G$ that is a monoid congruence (i.e., $w_1 \sim w_2$ and $w_1' \sim w_2' \implies w_1 w_1' \sim w_2 w_2'$). Do Theorems 0.2 and 0.3 still hold with M being the group $G$ and the homomorphism being a group homomorphism? Prove it?

(a) Define $N := [1_G] = \{g \in G \mid g \sim 1_G\}$, the equivalence class of the identity element. Prove that $N$ is a **normal subgroup** of $G$ ($N \trianglelefteq G$).

(b) Prove that for all $x, y \in G$,

$$x \sim y \iff x^{-1}y \in N \iff xN = yN.$$

Thus, the equivalence classes of $\sim$ are precisely the cosets of $N$.

(c) Prove that the quotient monoid $G/\sim$ is actually a **group**, and is canonically isomorphic to the standard quotient group $G/N$.

(d) Conversely, prove that for any normal subgroup $N \trianglelefteq G$, the relation $\sim_N$ defined by $x \sim_N y \iff x^{-1}y \in N$ is a congruence on $G$.

*Conclusion:* The theory of monoid congruences on a group $G$ coincides with the theory of normal subgroups and quotient groups. Every monoid congruence on a group is uniquely determined by a normal subgroup $N \trianglelefteq G$.

# References

1. Dummit, D. S., and Foote, R. M., *Abstract Algebra*.

2. Lang, S., *Algebra*.

Further references will be added as the notes develop.